Vivek Kaklotar

Module 1 – Overview of IT Industry

Q-1. What is a Program?

A program tells the computer what to do and how to do it**.**

Ex:- `print("Hello Vivek")`

LAB EXERCISE: Write a simple "Hello World" program in two different programming languages of your and syntax. choice. Compare the structure

1. Hello World in Python

```
print("Hello World")
```

Structure:

- No main() function required
- No header files
- No semicolon
- Direct execution

2. Hello World in C

```
#include <stdio.h>
int main()
{
printf("Hello World");
return 0;
}
```

Structure:

- Must include header file `#include <stdio.h>`
- Must define main() function
- Uses `printf()` to print
- Semicolon ; required
- return 0; ends the program

THEORY EXERCISE: Explain in your own words what a program is and how it functions.

- A program is a set of instructions that we give to a computer to do a specific task. These instructions are written in a programming language like C, Python, or Java. The computer cannot think on its own, so it follows the instructions given in the program step by step.

- In simple words, a program tells the computer what to do and how to do it.

- A program works in three main steps. First, it takes input from the user. Then it processes the input according to the instructions written in the program. Finally, it gives output as a result. This process is called the Input-Process-Output cycle.

- When we write a program, it is converted into machine language using a compiler or interpreter so that the computer can understand it. After that, the CPU executes the instructions and gives the result.

- So, a program is basically a way to communicate with the computer and make it perform useful tasks.

Q-2. What is Programming?

- Programming means writing instructions for a computer so that it can do a task. Since a computer cannot think on its own, we have to tell it exactly what to do.

- We use programming languages like C or Python to write these instructions. When we write code, the computer reads it, understands it, and then performs the task.

- In simple words, programming is the way we communicate with the computer and make it work according to our needs.

THEORY EXERCISE: What are the key steps involved in the programming process?

1) Problem Understanding
2) Planning the Solution
3) Writing the Code
4) Compiling and Running
5) Testing and Debugging
6) Maintenance

Q-3. Types of Programming Languages

1) Low-Level Languages
2) High-Level Languages
3) Middle-Level Languages

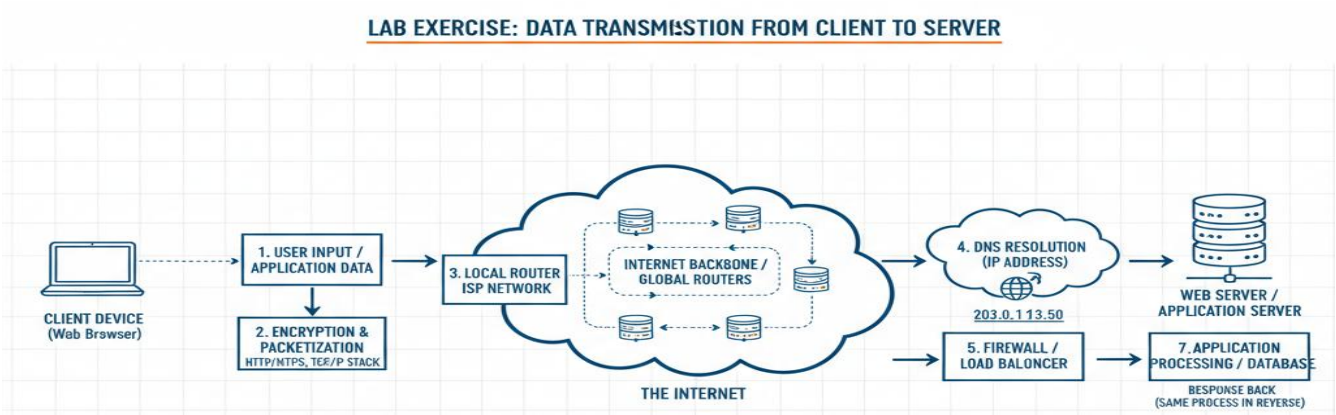
THEORY EXERCISE: What are the main differences between high-level and low-level programming languages?

High-Level Language	**Low-Level Language**
Easy to read and write	Difficult to read and write
Closer to human language	Closer to machine language
Portable (can run on different systems)	Machine dependent
Requires compiler/interpreter	Requires assembler
Less control over hardware	More control over hardware

Q-4. World Wide Web & How Internet Works

- World Wide Web
  - WWW is a collection of websites like Google, YouTube, Instagram, etc.
  - When we open a browser and type a website address (like www.google.com), we are using the World Wide Web.
- How Internet Works
  - The Internet is a global network of computers connected to each other.

LAB EXERCISE: Research and create a diagram of how data is transmitted from a client to a server over the internet.



THEORY EXERCISE: Describe the roles of the client and server in web communication.

- In web communication, the client and server work together to exchange information over the Internet.

- Role of the Client

  The client is the device or application that requests information.
  It is usually a web browser like Chrome or Edge running on a computer or mobile phone.
- Role of the Server

  The server is a powerful computer that stores websites, data, and applications.

Q-5. Network Layers on Client and Server

- In communication over the Internet, both the client and the server use network layers to send and receive data. These layers follow a structured model like the TCP/IP model (or OSI model).
- Each layer has a specific role in data communication.

LAB EXERCISE: Design a simple HTTP client-server communication in any language.

THEORY EXERCISE: Explain the function of the TCP/IP model and its layers.

- The TCP/IP model is a networking model that explains how data is transmitted over the Internet. It divides communication into different layers, and each layer has a specific function. This layered approach makes data transfer organized and reliable.
- The TCP/IP model mainly has **four layers**:

- Application Layer
- Transport Layer
- Internet Layer
- Network Access Layer

Q-6. Client and Servers

- Client
A client is a device or software that requests services or information from a server.
In simple words, the client asks for data.
    - Examples:
- A web browser like Chrome
- A mobile app
- A computer requesting a website

- Server
A server is a computer or program that provides services or data to clients.
In simple words, the server gives the requested information.
    - Examples:
- Web server
- Database server
- File server

THEORY EXERCISE: Explain Client Server Communication

Client–Server communication is a method of communication in which one computer (client) requests services or information, and another computer (server) provides that service or information.

In simple words, the client asks for data and the server gives the data.

Q-7 Types of Internet Connections

- Internet connection ke different types hote hain, depending on speed, technology aur availability.

1. Dial-Up Connection
2. Broadband Connection
3. Fiber-Optic Internet
4. Satellite Internet
5. Mobile Data (4G/5G)

LAB EXERCISE: Research different types of internet connections (e.g., broadband, fiber, satellite) and list their pros and cons.

1. Dial-Up Connection

➢ **Pros**
- Low cost
- Available in remote areas
- Simple setup

➢ **Cons**
- Very slow speed
- Cannot use phone and internet at the same time
- Outdated technology

2. Broadband Connection.

➢ **Pros**
- Faster than dial-up
- Always connected
- Affordable

➢ **Cons**
- Speed may decrease during peak hours
- Depends on service provider quality

3. Fiber-Optic Internet

➢ **Pros**
- Very high speed
- More reliable connection
- Low latency

➢ **Cons**
- Expensive installation
- Not available everywhere

4. Satellite Internet

➢ **Pros**
- Available in rural and remote areas
- No need for underground cables

➢ **Cons**
- High latency
- Affected by bad weather
- Expensive plans

5. Mobile Data (4G/5G)

➢ **Pros**

- Portable
- Easy to access
- Fast speeds (especially 5G)

➢ **Cons**

- Limited data plans
- Speed depends on signal strength

THEORY EXERCISE: How does broadband differ from fiber-optic internet?

Broadband is a general term for high-speed internet that includes DSL, cable, and wireless connections. Fiber-optic internet is a specific type of broadband that uses light signals through fiber cables.

Broadband	**Fiber-Optic**
Uses copper cables or DSL	Uses fiber-optic cables
Moderate speed	Very high speed
More signal loss	Less signal loss
Lower cost	Higher cost

Q-8 Protocols

A **protocol** is a set of rules that define how data is transmitted over a network.

Examples:

- HTTP
- HTTPS
- FTP
- SMTP

LAB EXERCISE: Simulate HTTP and FTP requests using command line tools (e.g., curl).

THEORY EXERCISE: What are the differences between HTTP and HTTPS protocols?

HTTP and HTTPS are protocols used for communication between a client and a server.

➢ **HTTP (HyperText Transfer Protocol)**

- Used to transfer web pages.
- Data is sent in plain text.
- Less secure.
- Uses port 80.

➢ HTTPS (HyperText Transfer Protocol Secure)

- Secure version of HTTP.
- Data is encrypted using SSL/TLS.
- More secure for sensitive data.
- Uses port 443.

HTTP	HTTPS
Not secure	Secure
No encryption	Uses encryption
Port 80	Port 443
Data can be intercepted	Data is protected

Q-9 Application Security

- Application security means protecting software applications from hackers, attacks, and data theft. Since applications store important information like passwords, personal data, and bank details, it is very important to keep them secure.

LAB EXERCISE: Identify and explain three common application security vulnerabilities. Suggest possible solutions.

1. SQL Injection
- SQL Injection happens when an attacker enters harmful SQL code into an input field (like a login form) to access or change the database.

What can happen:

- Unauthorized access to data
- Deleting or changing database records
- Stealing sensitive information

How to prevent it:

- Use prepared statements (parameterized queries)
- Validate and check user input

- Limit database access permissions

2. Cross-Site Scripting (XSS)

- XSS happens when an attacker adds harmful scripts into a website. These scripts run in other users' browsers.

What can happen:

- Stealing cookies
- Taking control of user sessions
- Redirecting users to fake websites

How to prevent it:

- Clean and validate user input
- Escape special characters
- Use security policies like Content Security Policy (CSP)

3. Cross-Site Request Forgery (CSRF)

- CSRF tricks a logged-in user into doing something without their knowledge, like sending money or changing account settings.

What can happen:

- Unauthorized transactions
- Account changes
- Data misuse

How to prevent it:

- Use CSRF tokens
- Verify user requests
- Use secure cookies

THEORY EXERCISE: What is the role of encryption in securing applications?
- Encryption protects data by converting it into a secret format that cannot be read by unauthorized people.

Why encryption is important:

1. **Protects data while sending it over the internet**
   When we use HTTPS, the data between the user and the server is encrypted.
2. **Protects stored data**
   Passwords and important information are encrypted or hashed before saving in databases.
3. **Keeps information private**
   Only authorized users can access encrypted data.
4. **Prevents data tampering**
   Encryption helps ensure that data is not changed by attackers.

Q-10 Software Applications and Its Types

- Software applications are programs that help users perform different tasks on a computer or mobile device. Software is mainly divided into two types:

- System Software
- Application Software

LAB EXERCISE: Identify and classify 5 applications you use daily as either system software or application software.

- **5 Applications I Use Daily**

➢ 1. Windows Operating System
Type: System Software
- Windows is system software because it manages the computer hardware and allows other programs to run.

➢ 2. Google Chrome
Type: Application Software
- Google Chrome is used for browsing the internet, so it is application software.

➢ 3. Microsoft Word
Type: Application Software
- It is used to create documents, so it is application software.

➢ 4. Android Operating System
Type: System Software
- Android manages mobile hardware and runs applications, so it is system software.

➢ 5. WhatsApp
Type: Application Software
- WhatsApp is used for messaging and communication, so it is application software.

THEORY EXERCISE: What is the difference between system software and application software?

- System software and application software have different purposes in a computer system.

➢ System Software

- System software is responsible for managing the computer's hardware and basic operations. It acts as a bridge between hardware and application software.

Examples:

- Windows
- Linux
- macOS
- Device drivers

➢ Application Software

- Application software is designed to help users perform specific tasks such as writing documents, browsing the internet, or editing photos.

Examples:

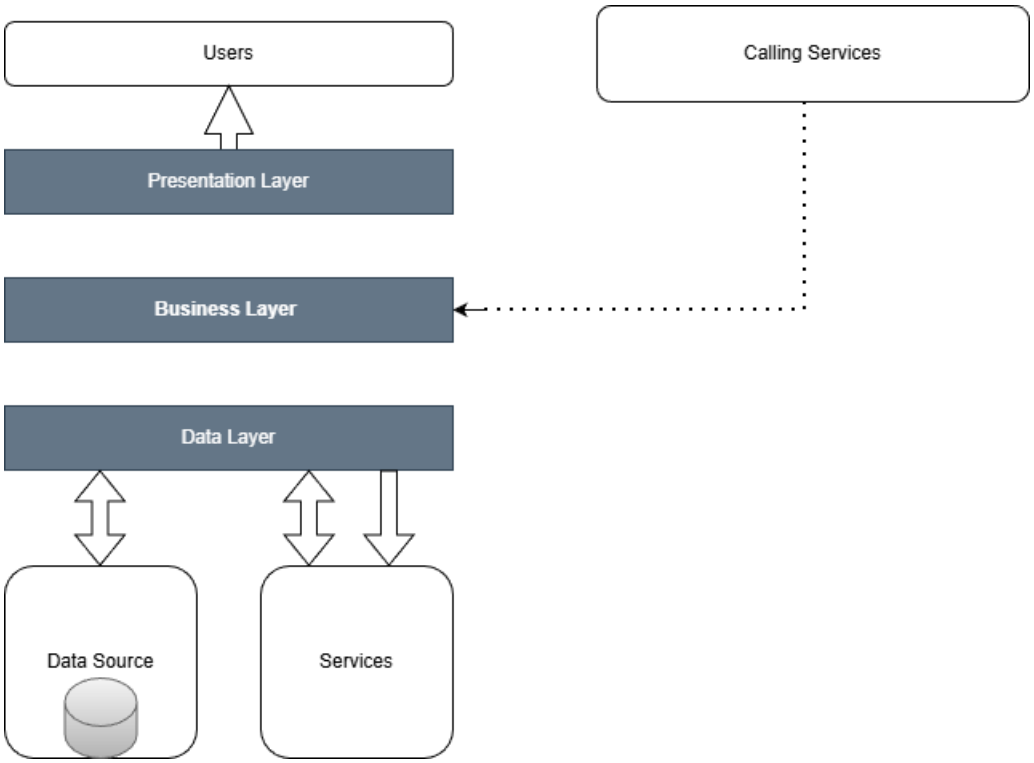
- Google Chrome
- MS Word
- Excel
- WhatsApp

System Software	Application Software
Controls hardware	Performs user tasks
Runs in background	Runs when user opens it
Necessary for system to work	Depends on system software
Example: Windows	Example: MS Word

Q-11 Software Architecture

- Software architecture refers to the overall structure of a software system. It defines how different parts of the system are organized and how they communicate with each other.

LAB EXERCISE: Design a basic three-tier software architecture diagram for a web application.



➢ Explanation of the Three Tiers

1 Presentation Layer (Client Layer)

- This is the user interface.
- It is what users see and interact with.
- Example: Web browser, mobile app UI.
- It sends user requests to the business layer.

2 Business Logic Layer (Application Layer)

- This layer processes the user's requests.
- It contains rules and logic of the application.

- Example: Login validation, payment processing.
- It communicates between presentation and data layer.

3 Data Layer (Database Layer)

- This layer stores and manages data.
- It includes databases like MySQL, MongoDB, etc.
- It sends data back to the business layer when requested.

THEORY EXERCISE: What is the significance of modularity in software architecture?

- Modularity means dividing a software system into smaller, independent modules. Each module performs a specific function.

➢ Why modularity is important:

1 Easy Maintenance
- If one module has a problem, it can be fixed without affecting the whole system.

2 Better Reusability
- Modules can be reused in other projects.

3 Easier Testing
- Each module can be tested separately.

4 Scalability
- New features can be added easily without changing the entire system.

5 Improved Teamwork
- Different developers can work on different modules at the same time.

Q-12 Layers in Software Architecture

- In software architecture, layers are used to separate different parts of a system based on their responsibilities. Each layer performs a specific function and communicates with other layers in a structured way.

- The most common layered architecture includes:

- Presentation Layer
- Business Logic Layer
- Data Access Layer

LAB EXERCISE: Create a case study on the functionality of the presentation, business logic, and data access layers of a given software system.

1 Presentation Layer (User Interface Layer)

Function:

This layer is responsible for interacting with the user.

In the Online Shopping System:

- Displays product lists
- Shows product details
- Provides login and registration forms
- Shows shopping cart and payment page

Example:

When a user searches for a product, the presentation layer collects the input and sends it to the business logic layer.

2 Business Logic Layer (Application Layer)

Function:

This layer processes user requests and applies business rules.

In the Online Shopping System:

- Validates login details
- Checks product availability
- Calculates total price
- Applies discounts
- Processes orders

Example:

When a user places an order, this layer checks stock availability and calculates the final bill.

3 Data Access Layer (Database Layer)

Function:

This layer handles communication with the database.

In the Online Shopping System:

- Stores customer details
- Stores product information
- Updates stock quantity
- Saves order history

Example:

When a product is ordered, this layer updates the product quantity in the database.

THEORY EXERCISE: Why are layers important in software architecture?
- Layers are important in software architecture because they divide the system into separate parts based on their functions.

1. Better Organization
2. Easy Maintenance
3. Improved Scalability
4. Better Testing
5. Reusability

Q-13 Software Environments

- A software environment is a setup where software applications are developed, tested, and used. Different environments are used at different stages of the software development process.

- The three main types of software environments are:

- Development Environment
- Testing Environment
- Production Environment

LAB EXERCISE: Explore different types of software environments (development, testing, production). Set up a basic environment in a virtual machine.

1 Development Environment

Purpose:

This is where developers write and modify the code.

Features:

- Code editors or IDEs (e.g., VS Code, IntelliJ)
- Programming languages installed (Python, Java, etc.)
- Debugging tools
- Local database

Example:

A developer writes and tests code on their local computer.

2 Testing Environment

Purpose:

This environment is used to test the software before releasing it.

Features:

- Similar to production setup
- Used by testers
- Used to find bugs and errors

Example:

Quality assurance (QA) team tests the application to check if everything works correctly.

3 Production Environment

Purpose:

This is the live environment where real users use the application.

Features:

- High security
- High performance
- Stable configuration

Example:

A live website used by customers.

➢ Setting Up a Basic Environment in a Virtual Machine

Step 1: Install Virtual Machine Software

Install software like:

- VirtualBox
- VMware

Step 2: Install an Operating System

Install an OS inside the virtual machine (e.g., Ubuntu or Windows).

Step 3: Install Required Tools

Inside the VM:

- Install programming language (Python/Java)
- Install database (MySQL)
- Install web server (Apache/Nginx)
- Install code editor

Step 4: Test the Setup

Create a simple program and run it to verify that the environment is working.

THEORY EXERCISE: Explain the importance of a development environment in software production.

1. Safe Experimentation
2. Error Detection
3. Improved Productivity
4. Consistency
5. Risk Reduction

Q-14 Source Code

- Source code is the original code written by a programmer using a programming language such as Python, Java, or C. It is human-readable and can be edited by developers.

LAB EXERCISE: Write and upload your first source code file to Github.

Step 1: Write a Simple Program

```
#include <stdio.h>
int main()
{
printf("hello im vivek.. \n");
printf("today is thrusday \n");
printf("im studying in tops \n");
return 0;
}
```

Save this file as:
first.c

Step 2: Create a GitHub Repository

1. Go to https://github.com
2. Click on **New Repository**
3. First.c
4. Click **Create Repository**

Step 3: Upload Using Git Commands

```
git init
git add .
git commit -m "module 1"
git branch -M main
```

git remote add origin https://github.com/Vivekkaklotar/vivek/tree/050200567ebee1ec46b5c391d4efff7665bb02b2/Assisgnment

```
git push -u origin main
```

THEORY EXERCISE: What is the difference between source code and machine code?

Source Code	Machine Code
Written by humans	Generated by compiler
Easy to read	Difficult to read
Written in programming language	Written in binary (0s and 1s)
Editable	Not easily editable

Q-15 GitHub and Introduction

- GitHub is an online platform used to store, manage, and share source code. It uses Git for version control.

LAB EXERCISE: Create a Github repository and document how to commit and push code changes.

1 Create Repository

- Login to GitHub
- Click “New Repository”
- Enter name and create

2 Clone Repository

```
git clone https://github.com/Vivekkaklotar/vivek.git
```

3 Add Files

```
git add .
```

4 Commit Changes

```
git commit -m "module 1"
```

5 Push Changes

```
git push origin main
```

THEORY EXERCISE: Why is version control important in software development?

Version control is important because it helps track changes in source code over time.

Importance:

1 Tracks Changes
- Developers can see who made changes and when.
2 Backup System
- Code is stored safely online.
3 Team Collaboration
- Multiple developers can work on the same project.
4 Restore Previous Versions
- Old versions of code can be recovered if needed.
5 Better Project Management
- Helps organize development work.

Q-16 Student Account in Github

- GitHub provides special benefits for students through the **GitHub Student Developer Pack**. It gives free access to tools, learning resources, and development software that help students improve their programming skills.

LAB EXERCISE: Create a student account on Github and collaborate on a small project with a classmate.

Part 1: Creating a GitHub Student Account

Step 1: Create a GitHub Account

1. Go to https://github.com
2. Click **Sign Up**
3. Enter your email, username, and password
4. Verify your email address

Step 2: Apply for GitHub Student Developer Pack

1. Go to: https://education.github.com/pack
2. Click on **Get Student Benefits**
3. Sign in with your GitHub account
4. Upload proof of student status (college ID or official email)

5. Submit the application

After approval, you will receive access to free tools and services.

Part 2: Collaborating on a Small Project

Step 1: Create a Repository

- Click **New Repository**
- Enter project name
- Click **Create Repository**

Step 2: Add a Classmate as Collaborator

1. Open your repository
2. Go to **Settings**
3. Click **Collaborators**
4. Enter your classmate's GitHub username
5. Send invitation

Step 3: Work Together

- Both can add code
- Commit changes
- Push updates
- Review each other's work

Basic Git Commands for Collaboration

```
git clone https://github.com/Vivekkaklotar/vivek.git
git add .
git commit -m "module 1"
git push origin main
```

THEORY EXERCISE: What are the benefits of using Github for students?

1. Free Developer Tools
2. Project Management
3. Collaboration
4. Version Control
5. Portfolio Building

Q-17 Types of Software

Software is divided into different types based on its function. The three main types are:

- System Software
- Application Software
- Utility Software

LAB EXERCISE: Create a list of software you use regularly and classify them into the following categories: system, application, and utility software.

Software I Use Regularly

➢ 1 Windows Operating System

Category: System Software
It manages the computer hardware and allows other programs to run.

➢ 2 Android Operating System

Category: System Software
It controls mobile hardware and runs applications.

➢ 3 Google Chrome

Category: Application Software
It is used to browse the internet.

➢ 4 Microsoft Word

Category: Application Software
It is used to create documents.

➢ 5 WhatsApp

Category: Application Software
It is used for messaging and communication.

➢ 6 Antivirus Software (e.g., Windows Defender)

Category: Utility Software
It protects the system from viruses and malware.

➢ 7 Disk Cleanup Tool

Category: Utility Software
It removes unnecessary files and improves system performance.

THEORY EXERCISE: What are the differences between open-source and proprietary software?

Open-Source Software	Proprietary Software
Source code available	Source code not available
Usually free	Usually paid
Can be modified	Cannot be modified
Community support	Official company support

Q-18 GIT and GITHUB Training

- Git is a version control system used to track changes in source code. GitHub is an online platform that hosts Git repositories and allows collaboration among developers.

LAB EXERCISE: Follow a GIT tutorial to practice cloning, branching, and merging repositories.

1 Cloning a Repository

Cloning means creating a copy of a remote repository on your local computer.

Step:

```
git clone https://github.com/Vivekkaklotar/vivek.git
```

This command downloads the repository to your system.

2 Creating and Switching to a Branch

Branching allows you to work on new features without affecting the main code.

Create a new branch:

```
git branch feature-branch
```

Switch to the new branch:

```
git checkout feature-branch
```

Or both in one command:

```
git checkout -b feature-branch
```

3 Making Changes and Committing

After editing files:

```
git add .
git commit -m "Added new feature"
```

4 Merging Branches

Switch back to the main branch:

```
git checkout main
```

Merge the feature branch:

```
git merge feature-branch
```

This combines the changes into the main branch.

5 Push Changes to GitHub

```
git push origin main
```

THEORY EXERCISE: How does GIT improve collaboration in a software development team?

- Git improves teamwork in many ways:
  1. Tracks Changes
  2. Multiple Developers Can Work Together
  3. Prevents Code Conflicts
  4. Easy Rollback
  5. Better Project Management

Q-19 Application Software.

- Application software refers to programs designed to help users perform specific tasks. Unlike system software, which manages the computer, application software helps users complete activities such as writing documents, browsing the internet, designing graphics, or managing data.

LAB EXERCISE: Write a report on the various types of application software and how they improve productivity.

Report on Types of Application Software

Application software can be divided into different types based on their purpose.

1 Word Processing Software

Examples: Microsoft Word, Google Docs

Purpose:

Used to create, edit, and format documents.

How It Improves Productivity:

- Easy document editing
- Spell check and grammar tools
- Quick formatting
- Saves time compared to handwritten work

2 Spreadsheet Software

Examples: Microsoft Excel, Google Sheets

Purpose:

Used for calculations, data analysis, and record keeping.

How It Improves Productivity:

- Automatic calculations
- Data organization
- Charts and graphs

- Financial and statistical analysis

3 Presentation Software

Examples: Microsoft PowerPoint, Google Slides

Purpose:

Used to create presentations.

How It Improves Productivity:

- Visual communication
- Easy slide design
- Saves time during meetings
- Improves explanation of ideas

4 Web Browsers

Examples: Google Chrome, Mozilla Firefox

Purpose:

Used to access websites and online information.

How It Improves Productivity:

- Quick access to information
- Online learning
- Email communication
- Research support

5 Communication Software

Examples: WhatsApp, Zoom, Microsoft Teams

Purpose:

Used for messaging, video calls, and meetings.

How It Improves Productivity:

- Instant communication
- Remote collaboration
- Online meetings
- Faster decision-making

6 Graphic Design and Multimedia Software

Examples: Adobe Photoshop, Canva

Purpose:

Used to create images, posters, and videos.

How It Improves Productivity:

- Professional design
- Marketing materials
- Creative content development

THEORY EXERCISE: What is the role of application software in businesses?

Application software plays a vital role in business operations. It helps organizations manage daily activities efficiently.

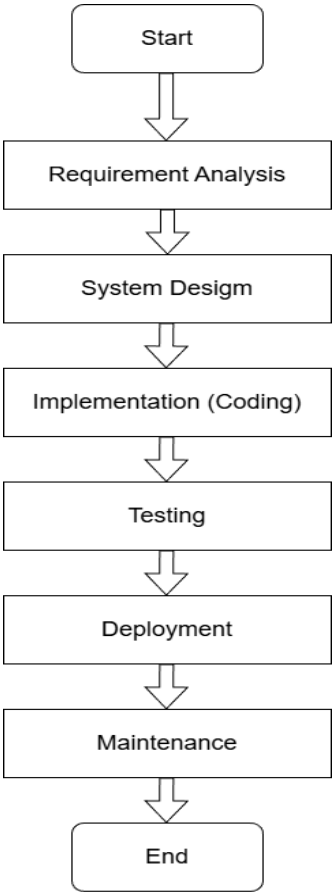
1 Data Management
Businesses use software to store and manage customer and financial data.
2 Communication
Email and video conferencing tools help employees communicate quickly.
3 Financial Management
Accounting software helps track income, expenses, and taxes.
4 Marketing and Sales
Businesses use design and CRM software to promote products and manage customers.
5 Decision Making
Data analysis tools help managers make informed decisions.

Q-20 Software Development Process

- The Software Development Process refers to the steps followed to develop a software application. These steps are organized into a structured method called the **Software Development Life Cycle (SDLC)**.

LAB EXERCISE: Create a flowchart representing the Software Development Life Cycle (SDLC).

THEORY EXERCISE: What are the main stages of the software development process?



1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Deployment
6. Maintenance

Q-21 Software Requirement

- Software requirements describe what a system should do and how it should perform. A requirement specification document clearly explains the features, functions, and constraints of the system before development begins.

LAB EXERCISE: Write a requirement specification for a simple library management system.

Requirement Specification

Library Management System (LMS)

1. Introduction

The Library Management System is a software application designed to manage the daily operations of a library. It helps in managing books, members, issuing and returning books, and maintaining records digitally.

2. Purpose

The purpose of this system is to:

- Reduce manual work.
- Maintain accurate records.
- Improve efficiency in managing library activities.

3. Scope

The system will:

- Store book details.
- Store member details.
- Allow issuing and returning books.
- Track due dates and fines.
- Generate reports.

4. Functional Requirements

These describe what the system must do.

1. The system shall allow the admin to add new books.
2. The system shall allow updating and deleting book records.

3. The system shall allow registering new members.
4. The system shall allow issuing books to members.
5. The system shall allow returning books.
6. The system shall calculate fines for late returns.
7. The system shall allow searching books by title, author, or ID.
8. The system shall generate reports of issued and available books.

5. Non-Functional Requirements

These describe how the system should perform.

1. The system should be user-friendly.
2. The system should respond within a few seconds.
3. The system should securely store data.
4. The system should support multiple users.
5. The system should be reliable and stable.

6. Hardware Requirements

- Computer or laptop
- Internet connection (for web-based system)

7. Software Requirements

- Operating System (Windows/Linux)
- Database (MySQL or similar)
- Programming Language (Java/Python/C# etc.)
- Web browser (if web-based)

THEORY EXERCISE: Why is the requirement analysis phase critical in software development?

The requirement analysis phase is very important because it defines the foundation of the software project.

1. Clear Understanding of User Needs
It ensures that developers understand exactly what the client wants.
2. Reduces Errors
Clear requirements prevent misunderstandings and mistakes later.
3. Saves Time and Cost
Fixing problems early is cheaper than fixing them after development.

4. Improves Software Quality
Well-defined requirements lead to better design and implementation.
5. Prevents Project Failure
If requirements are unclear, the project may not meet user expectations.

Q-22 Software Analysis
- Software analysis is the process of studying and understanding what a software system must do before designing and developing it. It focuses on identifying system requirements, user needs, and system functions.

LAB EXERCISE: Perform a functional analysis for an online shopping system.

Functional Analysis

Online Shopping System

Functional analysis describes what the system should do.

1 User Registration and Login

- The system should allow users to register with their details.
- The system should allow users to log in using username and password.
- The system should allow password recovery.

2 Product Management

- The system should display product categories.
- The system should allow users to search for products.
- The system should show product details (price, description, availability).
- The admin should be able to add, update, or delete products.

3 Shopping Cart

- The system should allow users to add products to the cart.
- The system should allow updating quantity.
- The system should calculate total price.

4 Order Processing

- The system should allow users to place orders.
- The system should generate an order ID.

- The system should store order history.
- The system should update product stock after purchase.

5 Payment System

- The system should support multiple payment methods (card, UPI, etc.).
- The system should confirm successful payment.
- The system should generate payment receipts.

6 Admin Functions

- Manage users.
- Manage products.
- View sales reports.
- Monitor orders.

THEORY EXERCISE: What is the role of software analysis in the development process?

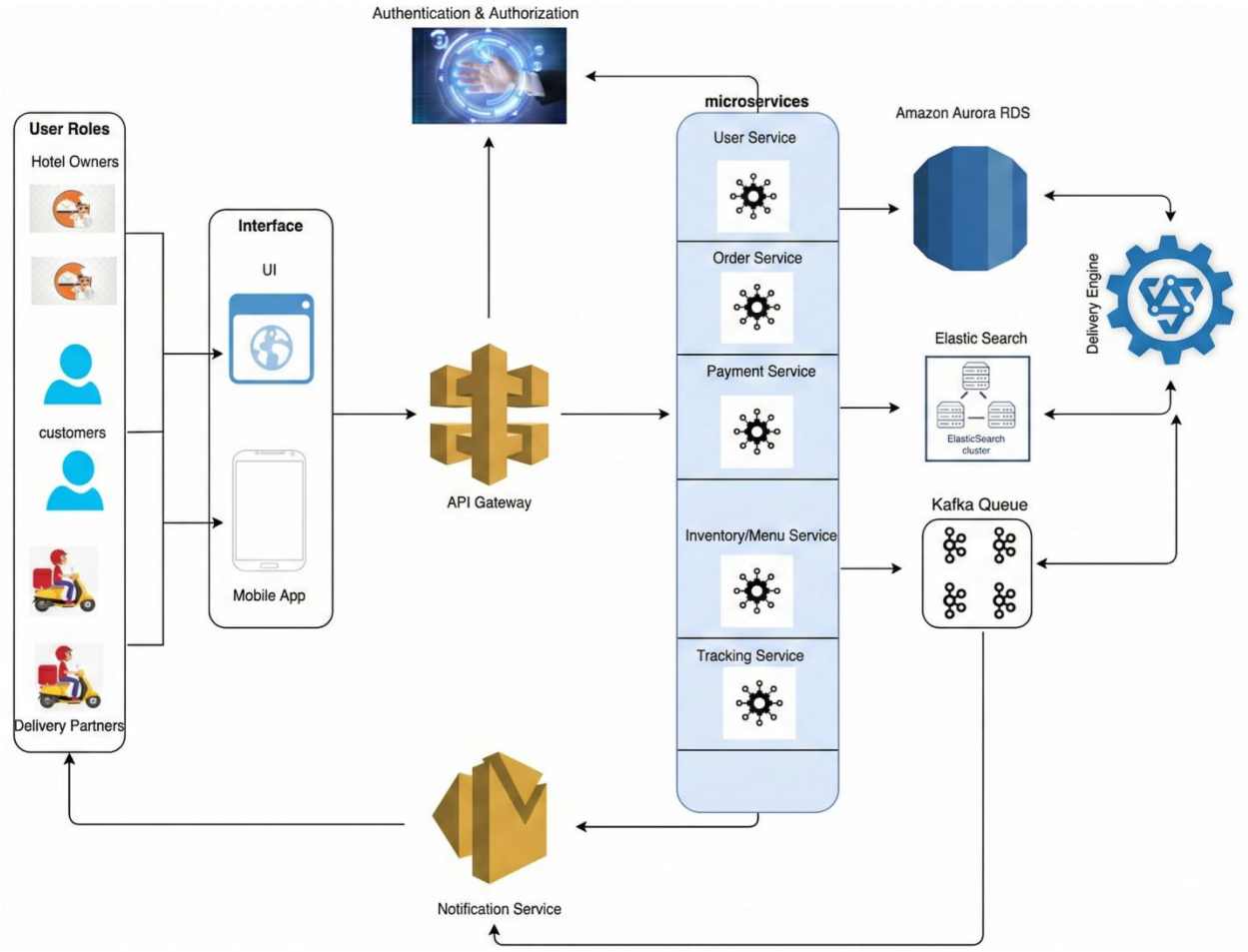
- Software analysis plays an important role in the development process because it defines what the system must do.

1 Understand User Needs
It ensures that developers clearly understand user requirements.
2 Define System Functions
It identifies all the functions and features of the system.
3 Reduce Errors
Proper analysis reduces mistakes during design and coding.
4 Improve Planning
It helps in better time and resource planning.
5 Prevent Project Failure
Clear analysis prevents misunderstandings and incomplete systems.

Q-22 System Design

- System design is the process of defining the structure, components, and architecture of a system. It explains how different parts of the system work together to achieve the required functionality.

LAB EXERCISE: Design a basic system architecture for a food delivery app.



(Fig: Basic System Architecture)

1 User Application (Customer App)

- Used by customers to browse restaurants.
- Place orders.
- Make payments.
- Track delivery.

2 Restaurant Panel

- Restaurants receive orders.
- Update menu and prices.
- Confirm or reject orders.

3 Delivery Partner App

- Delivery agents receive delivery requests.
- Track pickup and delivery location.
- Update delivery status.

4 Backend Server (Business Logic)

- Processes user requests.
- Manages authentication.
- Handles order management.
- Processes payments.
- Connects with database.

5 Database

- Stores user details.
- Stores restaurant information.
- Stores order history.
- Stores payment records.

6 Payment Gateway

- Processes online payments.
- Ensures secure transactions.

THEORY EXERCISE: What are the key elements of system design?

1. Architecture Design
2. Data Design
3. Interface Design
4. Component Design
5. Security Design
6. Scalability and Performance

Q-23 Software Testing

- Software testing is the process of checking whether a software application works correctly and meets the required specifications. It helps identify errors or bugs before the software is released to users.

LAB EXERCISE: Develop test cases for a simple calculator program.

THEORY EXERCISE: Why is software testing important?

- Software testing is important because it ensures that the software works correctly and meets user requirements.

1 Finds Errors
Testing helps identify and fix bugs before release.
2 Improves Quality
It ensures the software performs as expected.
3 Enhances Security
Testing helps detect security weaknesses.
4 Saves Cost
Fixing bugs early is cheaper than fixing them later.
5 Increases User Satisfaction
Reliable software builds trust among users.

Q-24 Maintenance

- Software maintenance is the process of updating and improving software after it has been released. It includes fixing bugs, improving performance, and adding new features.
- Maintenance is important because software must continue working properly even after deployment.

LAB EXERCISE: Document a real-world case where a software application required critical maintenance.

Case Study: WhatsApp Security Update

Background

WhatsApp is a widely used messaging application. In 2019, a security vulnerability was discovered in the app that allowed attackers to install spyware through a missed voice call.

Problem

- Hackers could exploit a weakness in the voice call feature.
- Users did not even need to answer the call.
- Sensitive information could be accessed without permission.

Maintenance Action Taken

- WhatsApp immediately released a security update.
- The vulnerability was fixed in the next version of the app.
- Users were advised to update the application.

Impact

- The update prevented further attacks.
- User data was protected.
- Trust in the application was maintained.

THEORY EXERCISE: What types of software maintenance are there?

1. Corrective Maintenance
2. Adaptive Maintenance
3. Perfective Maintenance
4. Preventive Maintenance

Q-25 Development

- Web applications and desktop applications are two different types of software applications. They differ in how they are developed, accessed, and used.

THEORY EXERCISE: What are the key differences between web and desktop applications?

Web Application	Desktop Application
Runs in browser	Installed on computer
Needs internet	Can work offline
Easy to access	Requires installation
Platform independent	Platform dependent
Auto updates	Manual updates

Q-27 Web Application

- A web application is a software program that runs on a web server and is accessed through a web browser using the internet.

- Unlike desktop applications, web applications do not need to be installed on a computer. Users can access them through browsers such as Chrome, Edge, or Firefox.

THEORY EXERCISE: What are the advantages of using web applications over desktop applications?

- Web applications have several advantages compared to desktop applications.

1 No Installation Required
Web applications run in a browser, so users do not need to install them.
2 Accessible from Anywhere
They can be accessed from any device with an internet connection.
3 Automatic Updates
Updates are done on the server side. Users always access the latest version.
4 Cross-Platform Compatibility
Web apps work on Windows, macOS, Linux, and mobile devices through a browser.
5 Lower Storage Usage
Web applications do not take up much storage space on the user’s device.
6 Easy Maintenance
Maintenance is easier because changes are made centrally on the server.

Q-28 Designing

- Designing in software development is the process of planning how a software system will look and work before coding begins. It defines the structure, layout, and behavior of the system.
- Designing helps developers understand how different parts of the system will interact with each other.

THEORY EXERCISE: What role does UI/UX design play in application development?

- UI (User Interface) and UX (User Experience) design play an important role in application development.

1 Improves User Experience
Good UX design ensures that the application is easy to use and understand.
2 Enhances Visual Appearance
UI design focuses on layout, colors, buttons, and overall look of the application.
3 Increases User Satisfaction
A well-designed application keeps users happy and engaged.
4 Reduces User Errors
Clear navigation and simple design reduce confusion.
5 Builds Brand Image
Attractive design improves the professional image of the application.

Q-29 Mobile Application

- A mobile application (mobile app) is a software program designed to run on mobile devices such as smartphones and tablets. These applications are created to perform specific tasks like messaging, shopping, gaming, or online banking.

- Mobile applications are usually downloaded and installed from app stores such as Google Play Store or Apple App Store.

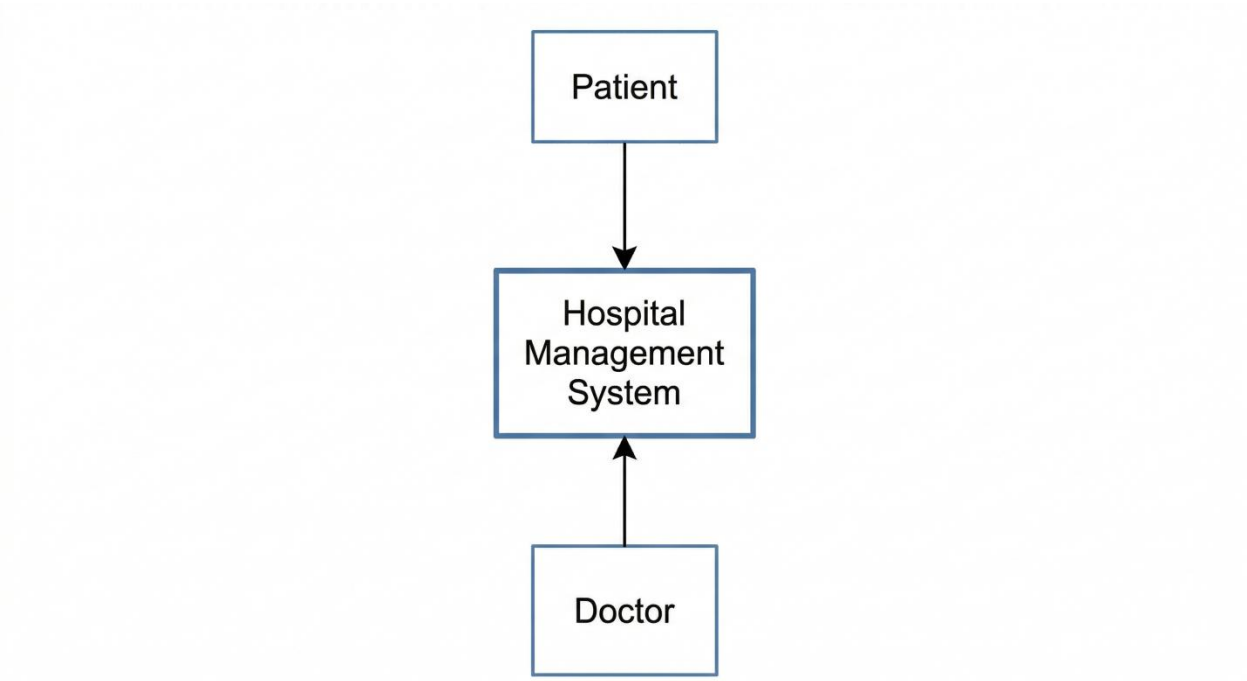
THEORY EXERCISE: What are the differences between native and hybrid mobile apps?

Native Apps	**Hybrid Apps**
Platform-specific	Works on multiple platforms
High performance	Moderate performance
Higher development cost	Lower development cost
Better user experience	Slightly limited performance

Q-30 DFD (Data Flow Diagram)

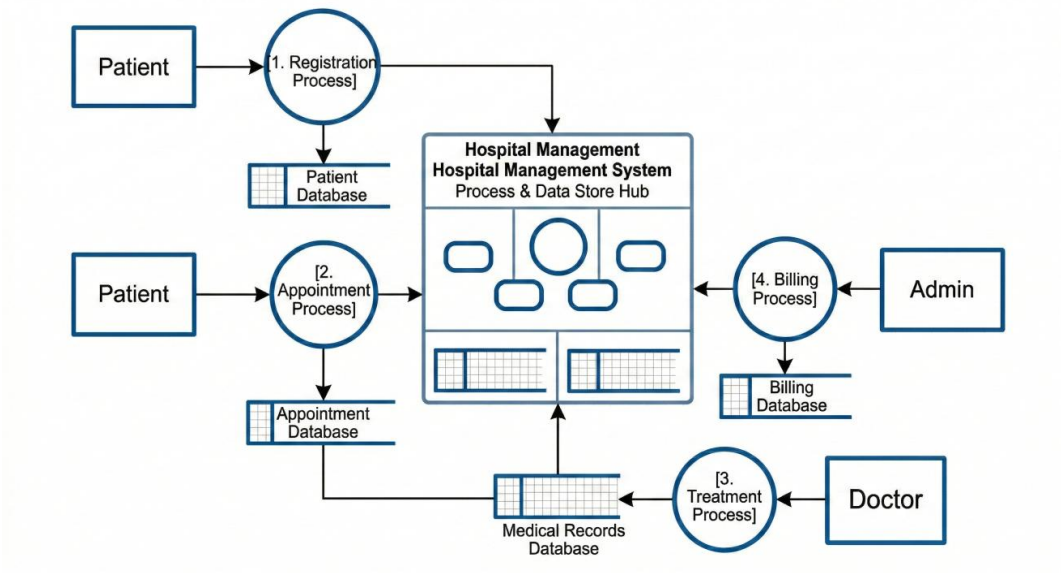
LAB EXERCISE: Create a DFD for a hospital management system.

- Level 0 DFD



(Fig : Context Diagram)

- Level 1 DFD



(Fig : Context Diagram)

THEORY EXERCISE: What is the significance of DFDs in system analysis?

DFDs are important because:

1 They show how data moves through the system.
2 They help understand system processes clearly.
3 They reduce confusion during development.
4 They help detect missing or incorrect data flow.
5 They improve communication between developers and clients.

Q-31 Desktop Application

LAB EXERCISE: Build a simple desktop calculator application using a GUI library.

THEORY EXERCISE: What are the pros and cons of desktop applications compared to web applications?

Pros and Cons of Desktop Applications Compared to Web Applications

Advantages of Desktop Applications

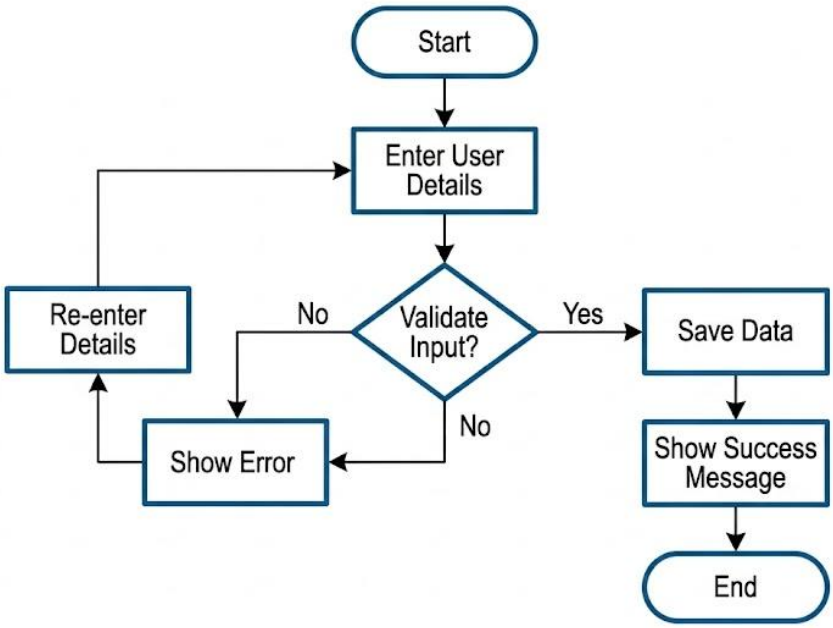
- Faster performance
- Can work offline
- Better system integration

Disadvantages

- Requires installation
- Platform dependent
- Manual updates required

Q-32 Flow Chart

LAB EXERCISE: Draw a flowchart representing the logic of a basic online registration system.



(Flowchart for Online Registration System)

THEORY EXERCISE: How do flowcharts help in programming and system design?

1 They show logic visually.
2 They make complex processes easier to understand.
3 They help identify errors early.
4 They improve communication among team members.
5 They help plan the program before coding.